Aatif Ali Khan

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Karachi, Pakistan — BS Computer Science, FAST NUCES — Expected Graduation: June 2027

Research Profile

Computer Science undergraduate at FAST NUCES with research interests in **Artificial Intelligence, Machine Learning, Foundation Models, Mechanistic Interpretability, and Graph Theory**. Currently conducting undergraduate research on genomic foundation models using **PyTorch and Sparse Autoencoders**, focusing on analyzing and interpreting learned representations. Strong foundation in **algorithms, AI, programming, and computational problem solving**, with additional experience in backend systems, databases, distributed computing, and cloud infrastructure. Seeking graduate research opportunities involving **machine learning, intelligent systems, and data driven scientific modeling**.

Education

FAST National University of Computer and Emerging Sciences (FAST NUCES) Karachi, Pakistan
Bachelor of Science in Computer Science Aug 2023 – May 2027 (Expected)
Relevant Coursework: Probability and Statistics, Calculus, Multi Variable Calculus, Linear Algebra, Artificial Intelligence, Graph Theory, Discrete Structures, Data Structures & Algorithms, Design & Analysis of Algorithms

Research & Technical Projects

Mechanistic Interpretability of Plant Genomic Foundation Models *2026 – 2027/Present*
Undergraduate Final Year Research Project — Python, PyTorch, Sparse Autoencoders

- Developing Sparse Autoencoder based methods to analyze internal representations of **AgroNT**, a 1B parameter genomic foundation model, with emphasis on interpretable features associated with drought resistance.
- Building PyTorch pipelines for model activation extraction, SAE training, hyperparameter evaluation, feature analysis, and representation visualization.
- Investigating causal relationships between learned representations and model predictions through **activation patching** and downstream genomic variant prioritization.

ExamScheduler AI, University Timetable and Exam Schedule Generator *Python, CSP, MRV, Backtracking, AI*

- Automated class, lab, and examination scheduling by modeling academic resources as a Constraint Satisfaction Problem to generate conflict free schedules.
- Resolved scheduling conflicts across courses, sections, teachers, rooms, and time slots by applying Minimum Remaining Values and backtracking search to produce consistent assignments.
- Supported real world scheduling workflows by implementing dataset import, configurable constraints, conflict validation, schedule export, and personalized student timetable and examination views.

AWS Auto Scaling & Load Balancing Infrastructure *2024*
Cloud Engineering Project — AWS, EC2, ALB, ASG, CloudWatch

- Automated EC2 provisioning using custom AMIs and Bash scripts and configured Application Load Balancing with Auto Scaling Groups for horizontally scalable workloads.
- Implemented CloudWatch monitoring and automated scaling policies and validated infrastructure behaviour through synthetic traffic and load testing.

Azure Cloud Based Scalable Network Infrastructure *2024*
Cloud Infrastructure Project — Microsoft Azure

- Designed a multi tier Azure VNet architecture with subnets, NAT Gateway, Network Security Groups, load balancing, CDN, and secure traffic routing for distributed workloads.
- Implemented network and DNS optimization strategies and evaluated workload performance under scalable infrastructure configurations.

Technical Experience

Decode Labs Karachi, Pakistan
Backend Development Intern June 2026 – July 2026

- Developed backend applications using **C#, ASP.NET Core, SQL Server, Entity Framework Core**, and RESTful API architecture.
- Implemented authentication, validation, dependency injection, structured error handling, and maintainable backend components using modern development practices.
- Collaborated through Git and GitHub based development workflows on project based backend tasks.

Technical Skills

AI / ML	Python, PyTorch, Transformers, Hugging Face, RAG, Sparse Autoencoders, Mechanistic Interpretability
Backend / Systems	C#, ASP.NET Core, REST APIs, Entity Framework Core, Microservices, Redis, RabbitMQ, PostgreSQL, SQL Server
Cloud / DevOps	AWS, Microsoft Azure, Docker, Bash, CloudWatch, Azure Monitor, Load Balancing, Auto Scaling
Programming	C, C++, C#, Python, JavaScript, SQL
CS Foundations	Data Structures & Algorithms, Distributed Systems, Operating Systems, Computer Networks, Databases, Software Engineering, OOP

Achievements & Certifications

- **Winner, Researchthon, AKUEB, 2022** – Led development of Hamsafar, securing first place among 50+ teams.
- **Top 5, Procom Coding Competition, 2023** – FAST NUCES.
- **Finalist, Developers Day Coding Competition, 2024**.
- **AWS Cloud Practitioner Essentials** – Amazon Web Services.
- **Microsoft Back End Developer Professional Certificate**.